

→ Block code that has a return value and it can be used again & again in a program.

→ Pascal convention.

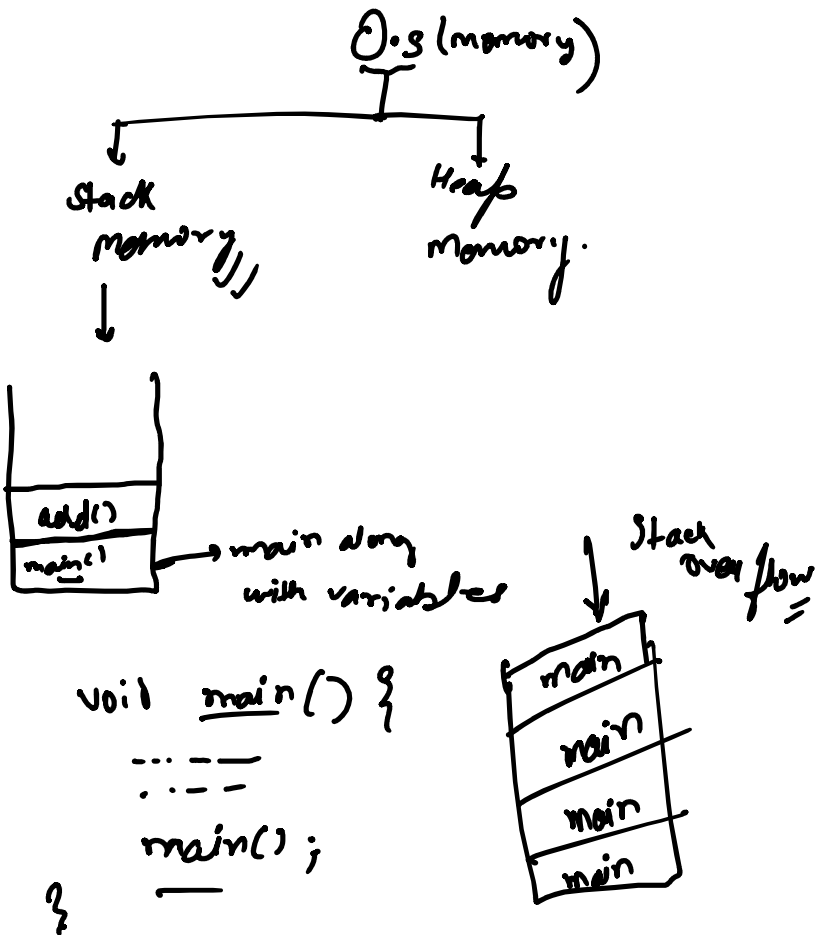
Public Class ClassName {

public class ClassName {
 public datatype functionName: {datatype variable_name, }
 }

- Private
- Profit Ad

Land Use Convention

2



form
name
Age

Hobbies: []

—
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Hobbies : - - - - -

Q. Write a function to return the sum of two numbers.

~~Public void sum () {~~ ^{int} ~~}~~ ✗

Public int sum(int a, int b) {
 return a+b;
 }
 ↓
 keyword.
 L → 2+5 = 7 (int) ✓

Public int add (long a, long b) {
 return ~~a+b;~~ (int) a+b;
 }

Q. Write a function to Convert a decimal number to binary number.

✓ Q. Binary to decimal. → How

Num = 23

2	23	1
2	11	1
2	5	1
2	2	0
1		

MSB

$$\begin{array}{r} 2 \overline{) 23} \\ \underline{12} \\ 11 \end{array}$$

$$\begin{array}{r} 2 \overline{) 11} \\ \underline{10} \\ 1 \end{array}$$

10111

$$\begin{array}{r} 1011 \\ 43210 \end{array}$$

$$\begin{aligned} 1 \times 2^0 &+ 1 \times 2^1 \\ &+ 1 \times 2^2 \\ &+ 0 \times 2^3 \\ &+ 1 \times 2^4 \\ &= 23 \end{aligned}$$

46 → 101110

2	46	0
2	23	1
2	11	1
2	5	1
2	2	0
1		

101110

→ 011101

→ 101110

2
↙
0, 1

```
int ans = 0;
while (num > 0) {
    int rem = num % 2; // rem
    ans = ans * 10 + rem;
    num = num / 2;
}
```

Q Prime factorization → write a function to print prime factors of a number.

Q Prime factorization - write a function to print prime factors of a number.

$$16 \rightarrow 2 \times 2 \times 2 \times 2$$

$$\begin{array}{r} 2 \overline{) 16} \\ \underline{2} \\ 2 \\ \underline{4} \\ 2 \\ \underline{2} \\ 0 \end{array}$$

$$28 = 2 \times 2 \times 7$$

$$46 = 2 \times 23$$

$$23$$

$$num = 1;$$

$$2$$

$$1/2, 2/2, \dots$$

$$2, 3, 4, \dots, 16$$

$$46 = 23$$

$$16 = 2 \times 2 \times 2 \times 2$$

$$16/2 = 8/2 = 4/2 = 2/2 = 1$$

$$1/3, 1/4, \dots, 1/16$$

$$28 \rightarrow 14 \rightarrow 7$$

$$2 \times 2 \times 7$$

$$\frac{5}{2}$$

$$30 \rightarrow 2 \times 3 \times 5$$

$$\frac{30}{2} \rightarrow 15 \rightarrow \frac{15}{2} \rightarrow \frac{15}{3} \rightarrow \frac{5}{3} \rightarrow \frac{5}{5} \rightarrow 1$$

$$\begin{array}{r} 2 \overline{) 30} \\ \underline{6} \\ 15 \\ \underline{15} \\ 0 \end{array}$$

$$37$$

$$2, 3, \dots, 37$$

2 2

37

16

$\frac{11}{2}$

2
2 2 2 2
4
2

```
for (int i=2; i<=n; i++) {
    while (n%i==0) {
        Sout (i+" ");
        n=n/i;
    }
}
```

$n \rightarrow \sqrt{n}$

$n=37$
✓

2 $\rightarrow$ 6 ... 37

$\sqrt{37} = 6$

if (n > 1) Sout(n);

$n=16$

$\sqrt{16} = 4$

2, 3, 4
 $\rightarrow$

$n=16$
 $\frac{16}{2} = 8$ $\frac{8}{2} = 4$ $\frac{4}{2} = 2$
2 2 2 2

$\frac{2}{2}$

$n=28$ $\sqrt{n} = 5$

$\rightarrow$ 2, 3, 4, 5

$\frac{28}{2} \rightarrow 0$ ✓ $\frac{14}{2} \rightarrow 0$ (7)
✓

2 2 7

$$\sqrt{n}$$

$$n = 36$$

$$\begin{array}{l} 1 \times 36 \quad \text{---} \\ 2 \times 18 \quad \text{---} \\ 3 \times 12 \quad \text{---} \\ 4 \times 9 \quad \text{---} \\ 6 \times 6 \quad \text{---} \\ \hline 9 \times 4 \quad \text{---} \end{array}$$

$$\sqrt{n}$$

$$n = 28 \rightarrow 2, 3, 4, 5$$

$$2 \quad 2$$

$$\frac{28}{2} \rightarrow \frac{14}{2} \rightarrow \frac{7}{2} \times$$

$$\frac{7}{3} \times \frac{7}{4} \times \frac{7}{5} \quad (7)$$

$$n = 16$$

$$\begin{array}{c} 2 \quad 2 \quad 2 \quad 2 \\ n = 1 \\ \sqrt{n} \rightarrow (4) \checkmark \end{array}$$

$$\boxed{n = \text{num}};$$

Prime number

$$\text{or } i$$

$$\text{if } (n > 1) \text{ print } (n);$$

$$\text{for } (i = 2 \rightarrow i \leq \sqrt{n} ; i++) \{$$

$$i \leq \sqrt{n}$$

Sq. box side,

$$i * i \leq n$$

H.W

✓ → Write a function to return the product of two numbers.

✓ → Pascal triangle print. (use function).

✓ → Write a function, if given a number return true if the number is an Armstrong number, otherwise false.

✓ → write a program to print all the prime numbers from 1 to N.